

DIABETO: A 360-DEGREE Diabetes-Management Solution



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"My parents have been living with type-2 diabetes for over 20 years. About four years ago, my father's diabetes file was misplaced by his hospital and he had to undergo all tests again. That was when we thought, we can actually solve this problem with the help of technology, and we came up with Diabeto." These are the words of Shreekanth Pawar, chief executive officer, Diabeto Medtech India Pvt Ltd, on being asked about what prompted the invention of Diabeto.

So what did they do?

They came up with a solution for diabetes management—a package to test blood-glucose levels, analyse test results, and get expert opinion and care, all from the comfort of your home. Read on to find out Diabeto's journey from inception to fruition.

Plug into your glucometer, pair with your smartphone and you are good to go

It is usual for a person suffering from diabetes to regularly check the blood sugar reading using a glucometer. Diabeto is a device that can be attached to a glucometer, simply by plugging its knob into the glucometer's jack. You can then pair the device to your smartphone via Bluetooth.

The smartphone app for Diabeto, supporting both iOS and Android, takes care of the rest.

A simple-outside, complex-inside design

It is important to have a device that gives the user not just an accurate experience but a happy one, too. With this in mind, the team

at Diabeto designed this piece of hardware that could easily be carried around and one that fits into the palm. In the design of a small bird, it symbolises freedom and comes in blue and pink colour variants.

Fitting everything into this miniature architecture and get it running, along with complying with all prevailing electronic and medical norms, offered enough roadblocks. It took the team three years to complete it, but they did it in style. They had to manufacture in China finally, as the quality of the product in India was sub-standard and the time, effort and money involved was simply making the whole process tedious.

The birdie contains a universal serial bus port, battery, Atmega microcontroller, Bluetooth Low Energy module and single-pole, double-throw switch. The switch shifts between the glucometer connection and the software universal asynchronous receiver/transmitter for respective functions.

How it works

On establishing a connection, the readings, time and date stamp from the glucometer are transferred to the app, while at the same time also getting stored in the cloud server, saving the trouble of entering the readings manually. The app then asks for details about meals (you could even upload a picture of the meal), wellness factors, insulin units taken and comments, which you can manually input. This additional data gets stored along with the readings and you can add details to past readings as well.

The next task is analysis. You can do this using a smartphone app or Web app. The app analyses all data stored and gives an insight into how your health is and what you need to do next. You can track your mood and food, measure physical activity, count calories from carbony-

Easy to use, good to see

